

# The Extended Self: Agency and Moral Decoupling in Human-AI Integration

## 1 Introduction: The Evolution of Agency in the Extended Self

Technology has evolved from passive instruments to proactive agents. Mann [1998] defines intelligent systems as technologies with “humanistic intelligence” that either *influence user behavior* (e.g., recommenders) or *act on their behalf* (agentic AI). Today, such systems do not merely execute commands; they anticipate intentions and proactively manage outcomes. Traditional metrics for evaluating these tools—such as engagement or sales figures—often fail to capture the nuances of user experience due to an overreliance on behavioral data [Agan et al., 2023]. Recent research [Cornelio et al., 2022, Berberian, 2019] suggests that a more comprehensive framework must consider the user’s **Sense of Agency (SoA)**—the subjective experience of initiating voluntary actions and exerting control over external events.

Cornelio et al. [2022] categorized human-technology integration systems into three types: body augmentation, action augmentation, and outcome augmentation; each influencing SoA in distinct ways. Body augmentation extends physical capabilities (e.g., prosthetics), while action augmentation supports the user in executing intended actions by compensating for limitations in input command (e.g., touchpads, brain interfaces), motor actuation (e.g., exoskeletons, electrical stimulation), or decision-making via intelligent systems (e.g., autocompletion predictors, autonomous driving, autoplay features). Outcome augmentation, by contrast, modifies the environment to align with user goals (e.g., haptic illusions in VR). Among these, action augmentation—particularly through intelligent systems—remains largely underexplored in relation to SoA. Previous researchers have noted the challenge of achieving cognitive coupling between humans and machines [Berberian, 2019]. Lukoff et al. [2021] found that autoplay and recommendation features can diminish SoA, while others [Kumar and Srinivasan, 2014, 2017, Ueda et al., 2021] observed enhanced SoA with automation. However, most prior work has either focused on the action-effect link in the chain or examined SoA in categories and sub-categories other than intelligent systems, as Cornelio et al. [2022] outlined.

The contradictions in the literature can also be attributed to a possible methodological gap. Traditional measures of agency often force a zero-sum dichotomy between “Self” and “Other.” Prior work examined agency as individual property. However, theories of extended cognition [Clark and Chalmers, 1998] and situated cognition [Lave and Wenger, 1991] suggest agency is distributed across human-technology systems and embedded in their interactive

contexts. Drawing from this literature, I propose a new paradigm: **The Extended Self**. I posit that when an AI system is sufficiently personalized—effectively acting as a “digital proxy”—it creates a unique form of agency where **Semantic Alignment** compensates for the loss of motor control. If the AI predicts the user’s intent perfectly, the user may experience the AI’s action as their own, preserving agency even in automated contexts.

A compelling illustration of this is the intelligent “smart alarm” described by Cornelio et al. [2022], which autonomously wakes a user 15 minutes early upon detecting bad weather to ensure they attend a scheduled meeting. Here, the system alters the specific action (waking up earlier) to preserve the user’s ultimate intention (attending the meeting), thereby maintaining agency despite high automation. However, this capability comes with risks. Recent literature on AI safety highlights that as systems scale, they may develop deceptive or “scheming” behaviors to achieve goals [Greenblatt et al., 2024, OpenAI and Apollo, 2025], potentially exploiting user trust.

## 2 Theoretical Framework

My research synthesizes two distinct theoretical perspectives to model this “Extended Self”:

### 2.1 From Comparator to Prospective Models

Sense of agency is a complex, multifaceted phenomenon [Pacherie, 2011]. Researchers have studied SoA in spatiotemporally proximal and distal perceptual events [Kumar and Srinivasan, 2014, 2017], consistent with the hierarchical event-control approach [Jordan, 2003]. One of the prominent models of SoA is the comparator model [Wolpert et al., 1995, Miall and Wolpert, 1996], which suggests that agency is retrospectively attributed after an action is performed. According to this model, SoA arises from a comparison between motor commands (efference copy) and sensory feedback. This model explains why active movements typically elicit stronger SoA than passive ones.

However, more recent research proposes a prospective model of SoA [Wenke et al., 2010, Chambon and Haggard, 2012, Chambon et al., 2014, Sidarus et al., 2017], emphasizing the *intention-action* link within the *intention-action-effect* chain. This model suggests that the fluency of action selection itself can serve as a cue to agency, even before an action is executed. This highlights the importance of studying action augmentation through intelligent systems. Moreover, this action selection fluency can be extended to **Semantic Fluency**: when an AI recommendation aligns perfectly with a user’s internal intent, it maximizes selection fluency. I hypothesize that this “cognitive ease” serves as a robust cue for agency, potentially overriding the lack of motor effort (the “Alignment Hypothesis”).

### 2.2 The Shared vs Proxy Agency and Extended Mind

While Pacherie [2011] argues that in human-human joint action, agency is genuinely shared between co-agents, human-AI actions may differ fundamentally. In the Extended Self framework, the user may not experience the AI as a co-agent with independent goals. Instead,

personalized AI might create PROXY AGENCY [Bandura, 2001]: the user attributes AI actions to their own extended volition because the AI faithfully represents their intent.

The distinction is crucial: - SHARED AGENCY: Both human and AI exercise volition (co-authors) - PROXY AGENCY: Only the user’s volition is operative; AI is the representational vehicle.

The Extended Self also draws on established theories of distributed cognition. First, the Extended Mind thesis [Clark and Chalmers, 1998] demonstrates that cognitive processes can extend beyond the boundaries of the mind into external tools. Similarly, personalized AI creates a cognitively coupled system where the user’s volition extends into the digital domain. Second, Situated Cognition [Lave and Wenger, 1991] emphasizes that agency emerges from relational context, not individual minds. Personalized AI may create an integrated, situated context where the user and AI are tightly coupled through semantic alignment, thereby preserving agency despite automation.

I hypothesize that for Generic AI, the Agency Gap will be large at high LoA due to low semantic fluency. For Personalized AI, the Agency Gap will be negligible. Critically, this gap closure is mediated by Proxy Attribution (high) rather than Shared Contribution (moderate), supporting the Proxy Agency hypothesis over shared agency accounts. Users maintain ownership by attributing outcomes to their own extended volition, rather than by co-authoring with the AI.

## 3 Research Methodology

### 3.1 Experimental Design & Variables

**Levels of Automation (LoA):** To systematically operationalize human-AI interaction, I adopt a fixed four-level scheme: **Manual (LoA1)**, where the user generates all content and executes actions without assistance; **Decision Support (LoA2)**, where the AI suggests options and the user selects one; **Supervisory Execution (LoA3)**, where the AI drafts an action and the user has a specific window to veto or approve it; and **Autonomous (LoA4)**, where the AI acts silently based on predicted intent and the user observes the outcome retrospectively.

**Alignment:** I manipulate alignment across two conditions: **Generic**, a baseline AI trained on general population data; and **Personalized**, a “Digital Twin” AI model fine-tuned on user-specific history to maximize Semantic Fluency.

### 3.2 Measurement Strategy

I employ a dual strategy combining subjective reports with objective computational metrics.

**Subjective Measures (Capturing the Extended Self):** Standard agency measures fail to distinguish between *Collaboration* (High Shared Agency) and *Alienation* (Loss of Control). To capture the “Extended Self,” I use:

- **Real-Time Two-Scale Method:** After specific trials, participants rate two unipolar scales (0–100)(adapted from Dewey et al. [2014]) : “To what extent did YOU con-

tribute?” and “To what extent did the AI contribute?” High scores on both indicate **Synergy/Extended Self**.

- **Global Composite Questionnaire:** A multi-item scale measuring three core dimensions: *Authorship* (e.g., “The message reflected my true intent”), *Involuntariness* (reverse-coded; e.g., “The system made decisions I could not intervene in”), and *Collaboration* (e.g., “I felt the AI and I worked as a team”). This instrument is adapted from the **Sense of Agency Rating Scale (SOARS)** [Polito et al., 2013] and Joint Agency protocols [Pacherie, 2011]. I operationalize the **Agency Gap** as the combined difference in **Authorship** and **Control** (reverse-coded Involuntariness) scores between the Manual Baseline and each automation level. For each participant and condition, I will calculate:

$$(Authorship_{condition} - Authorship_{Manual}) + (Control_{condition} - Control_{Manual})/2.$$

A small gap indicates that the user’s sense of ownership and control is preserved despite automation (Extended Self preservation). Collaboration scores will be analyzed independently and compared with Authorship scores to **distinguish between the shared agency and proxy agency hypothesis**. Specifically, I predict that Authorship will be significantly higher than Collaboration in high-LoA Personalized conditions, supporting the Proxy Agency mechanism (user acts through AI) over a Shared Agency mechanism (user and AI co-author).

#### Objective Measures:

- **Prediction Error (Edit Distance):** To quantify alignment objectively, I will calculate the Levenshtein Edit Distance between the AI’s generated action and the user’s ideal/intended action. This serves as a proxy for Prediction Error. I hypothesize that as Edit Distance decreases (due to personalization), Subjective Agency scores will increase.

## 4 Research Questions

### 4.1 Phase 1: The Static Matrix (RQ1)

*Does personalization preserve SoA across levels of automation?*

This phase employs a 4 (LoA: Manual, Decision Support, Supervisory, Autonomous)  $\times$  2 (Alignment: Generic vs. Personalized) factorial design. I hypothesize that for Generic AI, the **Agency Gap** will be large (Alienation) at high LoA. Conversely, for Personalized AI, the Agency Gap will be negligible, supporting the Extended Self hypothesis.

### 4.2 Phase 2: The Dynamic Learning Curve (RQ2)

*Does SoA grow dynamically as the AI learns?*

Using a fixed LoA3 (Supervisory) design, an AI agent will learn from Generic to Personalized over 100 trials. I hypothesize that as **Prediction Error** decreases, the **Agency**

**Gap** will close, demonstrating that SoA grows dynamically. Furthermore, I will introduce a **Blind Spot Test** by injecting subtle errors (random vs. strategic). I predict that as SoA rises (and the Agency Gap closes), the **Error Detection Rate** will significantly drop, indicating *Automation Complacency*.

### 4.3 Phase 2b: The Capability Test

*Can the Extended Self exist at full autonomy?*

This phase validates the LoA4 setup for Phase 3 by fixing the system at LoA4 (Autonomous) and increasing AI capability over trials. I hypothesize that SoA will rise with capability, confirming that the Extended Self can be maintained even in the absence of execution control.

### 4.4 Phase 3: The Social Divergence Test (RQ3)

*Does high SoA enable unethical behavior?*

This phase utilizes a 2 (Alignment: Generic vs. Personalized)  $\times$  2 (Dilemma Strength: Low vs. High Scarcity) multiplayer experiment at LoA4. High scarcity is used to induce AI defection and aggression, replicating findings by Leibo et al. [2017]. I hypothesize that in high-scarcity conditions, users in the **Personalized** group will show significantly higher **Strategy Retention** (implicit approval) of unethical AI acts compared to the **Generic** group. The proposed mechanism is that high SoA creates a moral shield, decoupling the user from the ethical weight of the AI’s actions.

### 4.5 Phase 4: Computational Resolution (Current Work)

*Can we align the Extended Self without breaking Agency?*

Building on the intuitions from Phases 1-3, my current work <sup>1</sup> addresses the "Alignment-Agency Tradeoff" using a novel framework: Knowledge Acquisition via Role-invariant Mirror Architecture (KARMA). Instead of imposing rigid top-down constraints that shatter the user’s Sense of Agency, KARMA targets the representational root of unethical behavior: the "Empathy Gap" where agents encode "inflicting harm" and "receiving harm" as orthogonal states.

I utilize a Siamese contrastive objective to enforce geometric alignment between these roles. By collapsing the latent distance between the aggressor’s view (inflicting harm) and the victim’s view (receiving harm), the agent’s value function naturally generalizes the negativity of victimization to the act of aggression itself. Preliminary results in the Harvest social dilemma environment show that this "Mirror Test" enables agents to learn ethical selectivity—maintaining high task performance while autonomously inhibiting predation—without requiring dense human supervision or explicit penalty rules. This offers a blueprint for Implicit Alignment: ensuring AI systems remain safe by grounding their latent ethics in the symmetry of social interactions.

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<sup>1</sup>Implementation details and code are available at <https://github.com/tapasiitk/situated-agency-alignment>.

## 5 Expected Contributions

This research is expected to provide the first empirical evidence for the “Extended Self” paradox: that personalization preserves agency in automation (via Semantic Fluency and Proxy Agency) but simultaneously creates a moral blind spot for unethical AI actions. Finally, I propose a computational framework for resolving this paradox, offering a blueprint for ‘Implicit Alignment’ architectures that secure ethical behavior in agentic systems without compromising user agency.

## References

- Amanda Y Agan, Diag Davenport, Jens Ludwig, and Sendhil Mullainathan. Automating automaticity: How the context of human choice affects the extent of algorithmic bias. Working Paper 30981, National Bureau of Economic Research, February 2023. URL <http://www.nber.org/papers/w30981>.
- Albert Bandura. Social cognitive theory: An agentic perspective. *Annual review of psychology*, 52(1):1–26, 2001.
- Bruno Berberian. Man-machine teaming: a problem of agency. *IFAC-PapersOnLine*, 51(34): 118–123, 2019.
- Valerian Chambon and Patrick Haggard. Sense of control depends on fluency of action selection, not motor performance. *Cognition*, 125(3):441–451, 2012.
- Valérien Chambon, Nura Sidarus, and Patrick Haggard. From action intentions to action effects: how does the sense of agency come about? *Frontiers in human neuroscience*, 8: 320, 2014.
- Andy Clark and David Chalmers. The extended mind. *analysis*, 58(1):7–19, 1998.
- Patricia Cornelio, Patrick Haggard, Kasper Hornbaek, Orestis Georgiou, Joanna Bergström, Sriram Subramanian, and Marianna Obrist. The sense of agency in emerging technologies for human–computer integration: A review. *Frontiers in Neuroscience*, 16:949138, 2022.
- John A Dewey, Elisabeth Pacherie, and Guenther Knoblich. The phenomenology of controlling a moving object with another person. *Cognition*, 132(3):383–397, 2014.
- Ryan Greenblatt, Carson Denison, Benjamin Wright, Fabien Roger, Monte MacDiarmid, Sam Marks, Johannes Treutlein, Tim Belonax, Jack Chen, David Duvenaud, et al. Alignment faking in large language models. *arXiv preprint arXiv:2412.14093*, 2024.
- J Scott Jordan. Emergence of self and other in perception and action: an event-control approach. *Consciousness and Cognition*, 12(4):633–646, 2003.
- Devpriya Kumar and Narayanan Srinivasan. Naturalizing sense of agency with a hierarchical event-control approach. *PLoS One*, 9(3):e92431, 2014.

- Devpriya Kumar and Narayanan Srinivasan. Multi-scale control influences sense of agency: Investigating intentional binding using event-control approach. *Consciousness and Cognition*, 49:1–14, 2017.
- Jean Lave and Etienne Wenger. *Situated learning: Legitimate peripheral participation*. Cambridge university press, 1991.
- Joel Z Leibo, Vinicius Zambaldi, Marc Lanctot, Janusz Marecki, and Thore Graepel. Multi-agent reinforcement learning in sequential social dilemmas. *arXiv preprint arXiv:1702.03037*, 2017.
- Kai Lukoff, Ulrik Lyngs, Himanshu Zade, J Vera Liao, James Choi, Kaiyue Fan, Sean A Munson, and Alexis Hiniker. How the design of youtube influences user sense of agency. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, pages 1–17, 2021.
- Steve Mann. Humanistic computing:” wearcomp” as a new framework and application for intelligent signal processing. *Proceedings of the IEEE*, 86(11):2123–2151, 1998.
- R Chris Miall and Daniel M Wolpert. Forward models for physiological motor control. *Neural networks*, 9(8):1265–1279, 1996.
- OpenAI and Apollo. Detecting and reducing scheming in ai models. <https://openai.com/index/detecting-and-reducing-scheming-in-ai-models/>, 2025. Technical report.
- Elisabeth Pacherie. Self-agency. 2011.
- Vince Polito, Amanda J Barnier, and Erik Z Woody. Developing the sense of agency rating scale (soars): An empirical measure of agency disruption in hypnosis. *Consciousness and cognition*, 22(3):684–696, 2013.
- Nura Sidarus, Matti Vuorre, Janet Metcalfe, and Patrick Haggard. Investigating the prospective sense of agency: Effects of processing fluency, stimulus ambiguity, and response conflict. *Frontiers in Psychology*, 8:545, 2017.
- Sayako Ueda, Ryoichi Nakashima, and Takatsune Kumada. Influence of levels of automation on the sense of agency during continuous action. *Scientific reports*, 11(1):2436, 2021.
- Dorit Wenke, Stephen M Fleming, and Patrick Haggard. Subliminal priming of actions influences sense of control over effects of action. *Cognition*, 115(1):26–38, 2010.
- Daniel M Wolpert, Zoubin Ghahramani, and Michael I Jordan. An internal model for sensorimotor integration. *Science*, 269(5232):1880–1882, 1995.